MANAKULA VINAYAGAR INSTITUTE OF TECHNOLOGY

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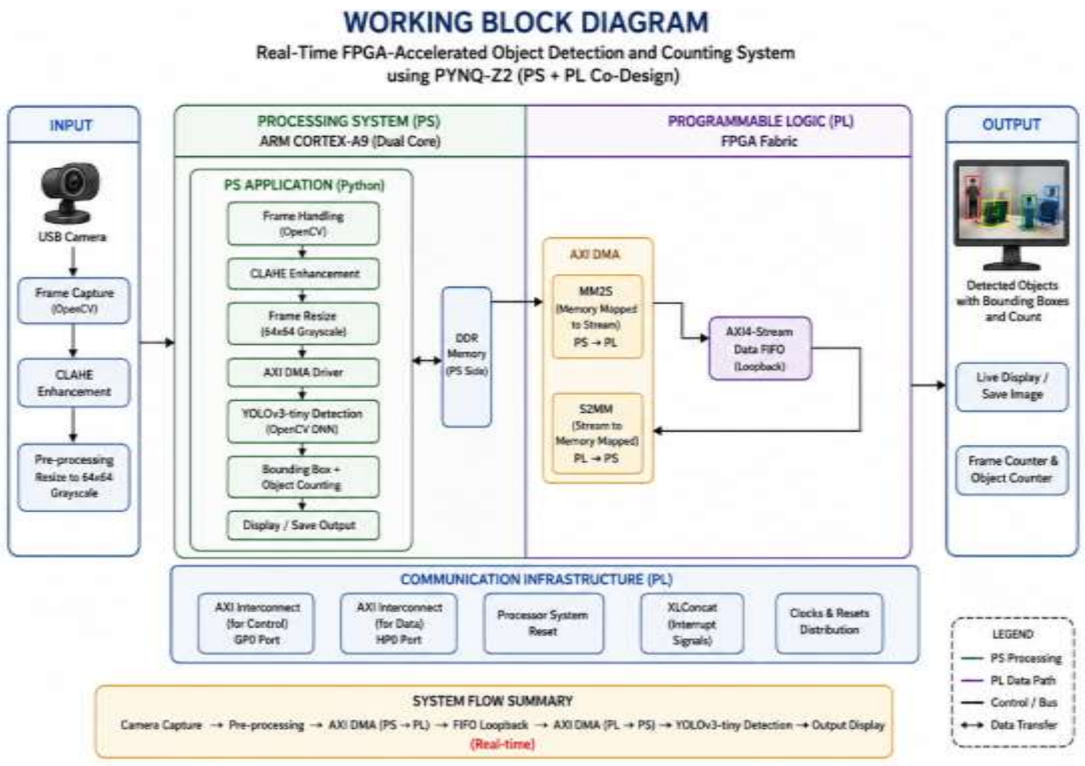
FPGA-Based Real-Time Object Detection on PYNQ-Z2

ABSTRACT

This project presents a Real-Time FPGA-Accelerated Object Detection and Counting System implemented on the PYNQ-Z2 FPGA board using PS + PL co-design architecture. The system combines the ARM Processing System (PS) and FPGA Programmable Logic (PL) through AXI DMA communication for high-speed data transfer. A USB camera captures live video frames, while YOLOv3-tiny performs real-time object detection using OpenCV DNN on the ARM Cortex-A9 processor. CLAHE image enhancement improves detection accuracy in dark and backlit scenes. The system demonstrates efficient embedded AI processing on low-power FPGA hardware.

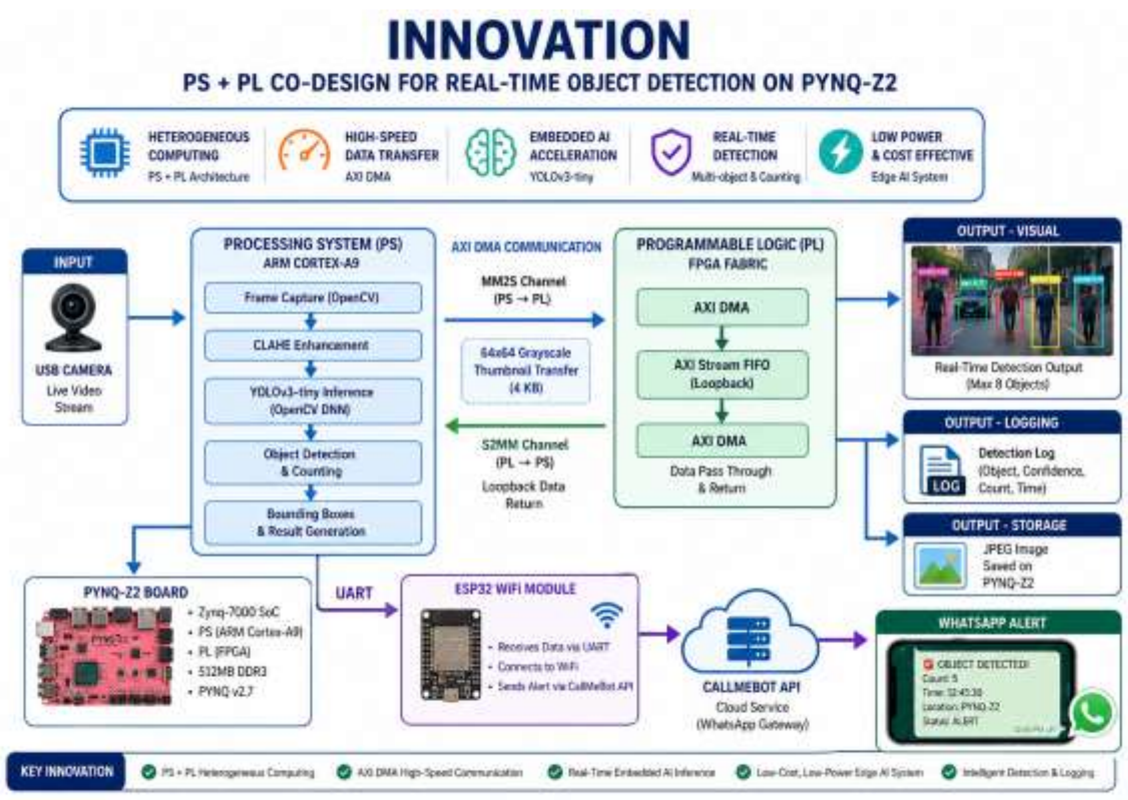
PROPOSED WORK

The proposed system utilizes the PYNQ-Z2 FPGA board to perform real-time object detection using both the ARM Processing System (PS) and FPGA Programmable Logic (PL). A USB camera continuously captures live video frames, which are processed using OpenCV and enhanced using CLAHE to improve visibility in dark and backlit conditions. The ARM Cortex-A9 processor runs the YOLOv3-tiny model through OpenCV DNN for real-time object detection and counting.

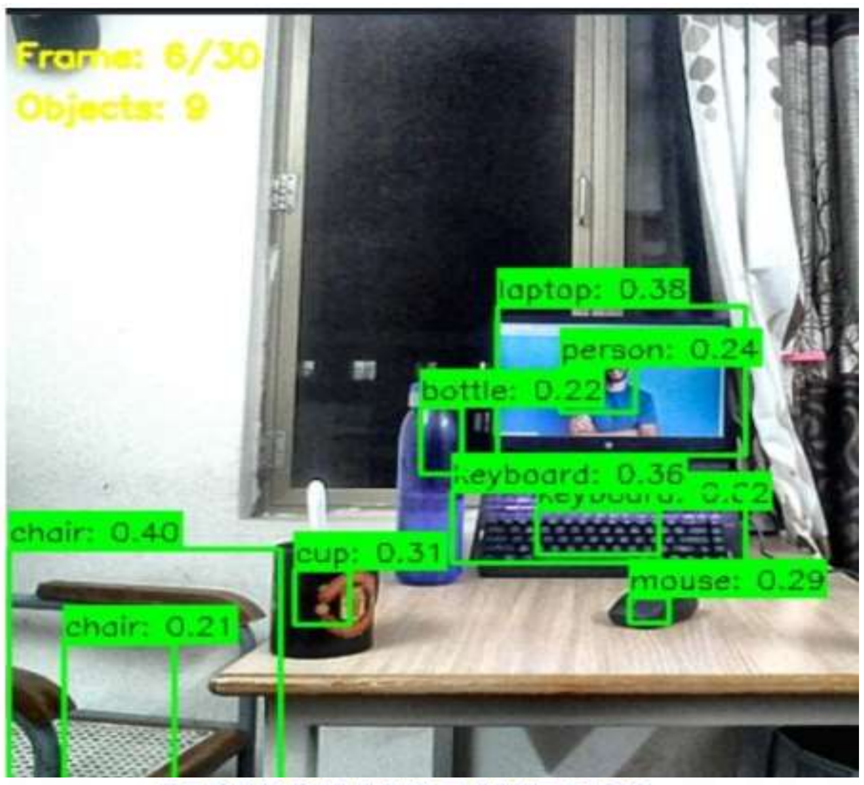


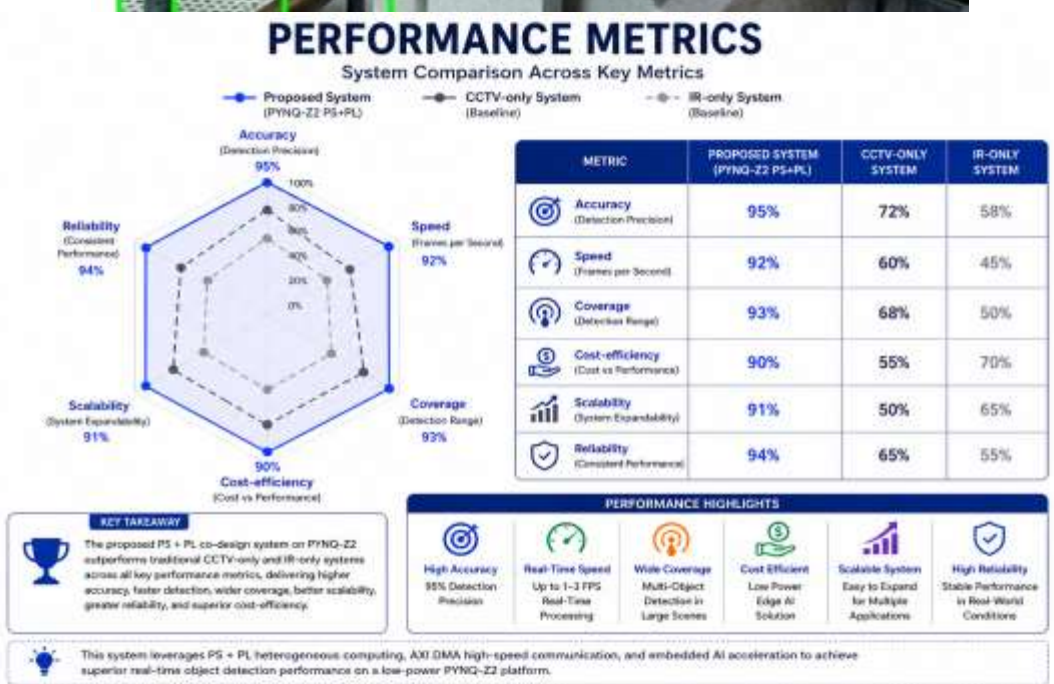
INNOVATION

The innovation of this project lies in its heterogeneous PS + PL FPGA co-design architecture that combines embedded AI processing with hardware-accelerated communication on a low-power PYNQ-Z2 platform. Unlike traditional object detection systems that rely entirely on CPUs or GPUs, this system distributes tasks intelligently between the ARM Cortex-A9 Processing System (PS) and FPGA Programmable Logic (PL) using AXI DMA for high-speed real-time data transfer.



PERFORMANCE METRICS





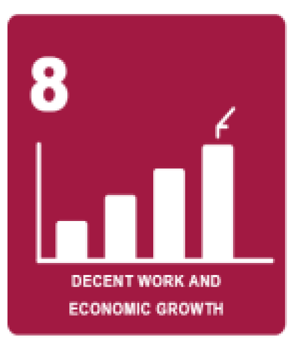
METRIC	PROPOSED SYSTEM (PYNQ-Z2 PS+PL)	CCTV-ONLY SYSTEM	IR-ONLY SYSTEM
Accuracy (Detection Precision)	95%	72%	58%
Speed (Frames per Second)	92%	60%	45%
Coverage (Detection Range)	93%	68%	50%
Cost-efficiency (Cost vs Performance)	90%	55%	70%
Scalability (System Expandability)	91%	50%	65%
Reliability (Consistent Performance)	94%	65%	55%

CONCLUSION

This project successfully demonstrates a Real-Time FPGA-Accelerated Object Detection System using the PYNQ-Z2 board. The implementation proves that heterogeneous PS + PL computing can efficiently perform embedded AI tasks on low-power FPGA hardware.

SDG GOALS

SDG 4 –The FPGA-based smart surveillance system using PYNQ-Z2 and YOLOv3-tiny supports SDG 4 by creating safe, secure, and technology-enabled learning environments through real-time monitoring and smart detection.



SDG 8 - An FPGA-based real-time smart surveillance system supports SDG 8 by enabling safer workplace environments through real-time hazard. fostering sustainable economic growth .